1. **Paper Title:** PlantDet: A Robust Multi-Model Ensemble Method Based on Deep Learning For Plant Disease Detection

   All Authors Name : Md Sakib Hossain Shovon; Shakrin Jahan Mozumder; O. Pal; M. Mridha; Nobuyoshi Asai; Jungpil Shin

   Publication year : 2022

   Objective: Develop robust ensemble for rice and betel leaf disease detection

   Methodology : Deep ensemble of InceptionResNetV2, EfficientNetV2L, Xception; data augmentation; Grad-CAM/Score-CAM

   Key Findings: Achieved 98.53% accuracy on rice, and ensemble surpassed base and other ensembles on betel leaf dataset

   Contribution to the field: First deep ensemble including betel leaf classes; strong benchmark for small, heterogeneous datasets

   Limitations: Only RGB modality; fusion is model-ensemble, not multimodal (image+sensor)

   Relevance : Direct betel leaf classifier; ensemble idea can be combined with multimodal fusion inputs

   URL or DOI: https://doi.org/10.1109/ACCESS.2023.3264835

2. **Paper Title:** Betel Leaf Diseases Classification using Machine Learning Algorithm: A Feasible Approach

   All Authors Name: Femi David; M. A. Mukunthan

   Publication year: 2024

   Objective: Classify multiple betel leaf diseases using classical ML

   Methodology : ROI-based segmentation; GLCM texture features; Extreme Learning Machine (ELM) classifier

Key Findings: On 1,047 images (5 diseases) achieved 97% accuracy

Contribution to the field: Strong classical pipeline specialized for betel leaf; shows ELM can rival deep models

Limitations: Handcrafted features; no deep learning; image-only (no fusion)

Relevance : Core baseline for betel leaf; can be extended with deep/fusion approaches

URL or DOI: https://doi.org/10.37934/araset.40.1.7486

3. **Paper title:** Betel Leaf Disease Classification Using Data Augmentation and Convolutional Neural Network

All Authors Name : A.S. Devi; S. Albert Antony Raj

Publication Year: 2023

Objective : Improve betel leaf disease classification with augmented data

Methodology : 1,089 field images (3 classes); heavy augmentation; simple CNN and VGG16

Key Findings: VGG16 with augmentation achieved 86.67% accuracy, outperforming simple CNN

Contribution to the field: Demonstrates impact of augmentation and transfer learning on real-field betel images

Limitations: Only 3 classes; modest accuracy; no multimodal fusion

Relevance: Early deep-learning study on betel leaf; candidate baseline for adding fusion

URL/DOI : rp-9788770040723.168

4. **Paper Title:** A comprehensive image dataset for accurate diagnosis of betel leaf diseases using artificial intelligence in plant pathology

   All Authors Name: Rashidul Hasan Hridoy; Md. Tarek Habib; Imran Mahmud; Aminul Haque; Md. Abdulla Al Mamun

   Publication Year: 2025

   Objective: Publish extensive betel leaf image dataset for AI diagnosis

   Methodology: 2,037 images (3 classes) collected 2020–2022; augmented to 10,185 images; natural field conditions

   Key Findings: Provides well-curated 10k+ image dataset; fills gap of reproducible betel dataset

   Contribution to the field: Major data resource for betel leaf DL; supports benchmarking and future fusion with sensors

   Limitations: Only 3 disease types; no non-image modalities; no new classifier

   Relevance: Foundational dataset for any betel leaf classification or multimodal fusion model

   URL or DOI: 10.1016/j.dib.2025.111564

5. **Paper Title :** CBAM Integrated Attention Driven Model For Betel Leaf Diseases Classification With Explainable AI

   All Authors Name: Sumaiya Tabassum; Md. Faysal Ahamed

   Publication year: 2025

   Objective: Build lightweight attention-CNN for betel leaf with explainable AI

   Methodology: Custom CBAM-CNN (~2.13M params); CBAM attention; Grad-CAM

for explainability; trained on 10,185-image betel dataset

Key findings: Achieved 95.58% accuracy, precision 97%, recall 94%, F1 95%; outperformed heavy pre-trained CNNs

Contribution to the field: Shows benefit of attention mechanisms and lightweight models specifically for betel leaf

Limitations: Still single-modality RGB; fusion is intra-image attention, not multimodal; arXiv preprint stage

Relevance: Directly addresses betel leaf; architecture suitable to extend with sensor/context fusion

URL/DOI: https://doi.org/10.48550/arXiv.2509.26484

6. **Paper title :** Deep Learning based Betelvine Leaf Disease Detection (Piper betle L.)

   All Authors Name: R. Kavitha Lakshmi; S. Nickolas

   Publication year: 2020

   Objective: Detect betelvine leaf diseases using DL

   Methodology: CNN-based deep learning (details referenced as DL betelvine detection)

   Key findings: Reported effective betelvine disease detection using DL, referenced as early betelvine CNN work

   Contribution to the field: Among the first dedicated Piper betle DL classifiers

   Limitations: Limited details in citation; dataset, architecture, and fusion not fully described

   Relevance: Early DL baseline for betelvine, informing later architectures

URL/DOI : 10.1109/ICCCA49541.2020.9250911

7. **Paper title:** Early Recognition of Betel Leaf Disease using Deep Learning with Depth-wise Separable Convolutions

All Authors name: Authors cited as Hridoy et al. in other works

Publication year:2021

Objective: Achieve fast, early betel leaf disease recognition

Methodology: Custom BLCNN with depth-wise separable convolutions; Swish activation

Key findings: BLCNN achieved 96.02% test accuracy on 1,031 images, outperforming standard CNN (89.53%)

Contribution to the field: Shows lightweight depth-wise CNNs effective for betel, enabling mobile deployment

Limitations: Uses only RGB images and three disease classes; no multimodal fusion

Relevance: Core betel leaf DL architecture; suitable base for adding external modalities

URL/DOI: 10.1109/TENSYMP52854.2021.9551009

8. **Paper title:** Improved vision-based diagnosis of multi-plant disease using an ensemble of deep learning methods

All Authors Name: Rashidul Hasan Hridoy; Arindra Dey Arni; Aminul Haque

Publication year: 2023

Objective : Develop hybrid Xception–ResNet50 ensemble for multiple crops including

Betel

Methodology: Stacking ensemble (SELT) of Xception and ResNet50 on MPLI dataset (black gram, betel, Malabar spinach, litchi)

Key findings: On 44,972 images (9 classes) achieved 99.20% accuracy; first joint model for four crops including betel

Contribution to the plant: Demonstrates high-accuracy cross-crop ensemble and introduces multi-plant dataset with betel

Limitations: Only RGB images; no multimodal fusion; dataset limited to four crops

Relevance: Provides strong multi-plant reference including betel; ensemble idea extendable to fused inputs

URL/DOI : http://doi.org/10.11591/ijece.v13i5.pp5109-5117

9. **Paper title:** Leaf image based plant disease identification using transfer learning and feature fusion

All Authors Name: Xijian Fan; Peng Luo; Yuen Mu; Rui Zhou; T. Tjahjadi; Yi Ren

Publication year: 2022

Objective: Improve plant disease recognition using deep + handcrafted feature fusion

Methodology: Transfer-learning CNN features + handcrafted textures; feature-level fusion; center loss

Key findings: Achieved up to 99.79% accuracy on apple/coffee datasets with fused features and center loss

Contribution to the field: Demonstrates feature-level fusion (deep + handcrafted) can significantly improve disease classification

Limitations: Not betel-specific; single-sensor RGB; no environmental fusion

Relevance: Fusion strategy (deep+texture+center loss) is directly portable to betel leaf images

URL/DOI: https://doi.org/10.1016/j.compag.2022.106892

10. **Paper title:** Plant disease detection using computational intelligence and image processing

    All Authors Name: Vibhor Kumar Vishnoi; Krishan Kumar; Brajesh Kumar

    Publication year: 30-08-2020

    Objective: Survey computational intelligence and image processing methods for plant disease detection

    Methodology: Review of computer vision, soft computing, modern feature extraction, and classification methods on leaf images

    Key findings: Summarizes merits/demerits of classical and modern approaches and identifies effective feature extraction techniques across crops

    Contribution to the field: Provides structured overview of image-based plant disease systems, highlighting role of advanced features and ML

    Limitations: Narrative review; no new model; limited coverage of recent deep multimodal fusion

Relevance: Establishes general design options for betel leaf systems (preprocessing, features, classifiers)

URL/DOI: https://link.springer.com/article/10.1007/s41348-020-00368-0

11. **Paper title:** Plant Disease Classification Using Image Processing Technique

    All Authors Name: Shahrul Fazly Man; Shafie Omar; Muhammad Imran Ahmad; Wan Mohd Faizal Wan Nik; Tan Shie Chow; Mohd Nazri Abu Bakar; Fadhilnor Abdullah; Asbhir Yuusuf

    Publication year: 03-06-2024

Objective: Rapidly classify specific plant diseases using an image-processing-based System

Methodology: Image preprocessing, segmentation, and ML classification targeting bacterial blight, Alternaria alternata, Cercospora leaf spot

Key findings: Achieved ~98–99% accuracy for three key diseases, demonstrating fast and effective categorization

Contribution: Provides a high-accuracy pipeline that can be adapted to other crops

Limitations: Focus on three diseases; no DL or multimodal fusion; not betel-specific

Relevance: Methods can be adapted as a lightweight alternative for betel farmers

URL/DOI : 10.58915/aset.v3i1.785

12. **Paper title:** Classification Of Plant Leaf Diseases Using Machine Learning And Image Preprocessing Techniques

    All Authors Name : Pushkar Sharma; P. Hans; Subhash Chand Gupta

    Publication year: 01-01-2020

Objective: Build AI-based automatic plant leaf disease detection and classification Pipeline

Methodology: Image collection, preprocessing, segmentation and ML-based Classification

Key findings: Demonstrates feasibility of automated classification pipeline to increase

agricultural productivity

Contribution: Early integrated pipeline combining image preprocessing with ML Classifiers

Limitations: Limited details on models and datasets; no DL or multimodal fusion; not Betel-specific

Relevance: General pipeline steps transferable to betel leaf workflows

URL/DOI : 10.1109/Confluence47617.2020.9057889

13. **Paper title:** Applying the transfer learning models on the dataset on the effect of diseases on Nagvel-betel (Piper betle) leaves

    All Authors Name: Milind Gayakwad; Rahul Joshi; Tulshihar Patil; Pratvina Talele; Rajendra Pawar; Priyanka Paygude

    Publication year: 2025

    Objective: Provide large diseased-betel dataset and test ViT transfer learning

    Methodology: 4,156 high-resolution diseased betel images (15 mapped disease categories); Vision Transformer models (e.g., ResNet, ConvNeXtB, MobileNet, InceptionV3 used in survey)

    Key findings: Classification accuracy with ViT-based models ranges 0.7–0.9 across disease categories

    Contribution: Supplies another large betel dataset with many diseases and demonstrates ViT robustness

    Limitations: Only diseased leaves; no healthy class; single-modality RGB; no multimodal fusion

    Relevance: Key Piper betle dataset; ideal for advanced CNN/ViT and potential multimodal fusion

    URL/DOI: https://doi.org/10.1016/j.dib.2025.111987

14. **Paper title:** Deep Learning in Multimodal Fusion for Sustainable Plant Care: A Comprehensive Review

    All Authors Name: Zhi-Xiang Yang; Yusi Li; Rui-Feng Wang; Pingfan Hu; Wen-Hao Su

    Publication year: 2025

    Objective: Review DL-based multimodal fusion across plant care tasks

    Methodology: Survey of DL multimodal fusion frameworks, tasks (disease, yield, weeds), and sustainability issues

    Key findings: Highlights success of multisource fusion (RGB, multispectral, sensors) and challenges (heterogeneity, real-time)

    Contribution: Provides a conceptual map of multimodal fusion for plant disease diagnosis

    Limitations: Review only; no betel-specific experiments

    Relevance: Guides design of multimodal betel leaf systems (e.g., fusion of leaf images + IoT in betel farms)

    URL/DOI : https://doi.org/10.3390/su17125255

15. **Paper title:** Deep Learning for Plant Identification and Disease Classification from Leaf Images: Multi-prediction Approaches

    All Authors Name: Jianping Yao; Son N. Tran; Saurabh Garg; Samantha Sawyer

    Publication year: 2023

    Objective: Compare multi-prediction DL strategies for plant ID and disease

    Methodology: GSMo-CNN (Generalised Stacking Multi-output CNN) with InceptionV3, multi-output learning

Key findings: Shows single multi-output model can match/beat separate models; InceptionV3 best backbone

Contributions: Provides taxonomy of multi-task/label/output strategies and a strong GSMo-CNN design

Limitations: No betel data; no non-image fusion

Relevance: Multi-prediction design can be adapted to jointly predict betel disease type + severity or treatment

URL/DOI : https://doi.org/10.48550/arXiv.2310.16273

16. **Paper title:** Plant Leaf Disease Detection, Classification, and Diagnosis Using Computer Vision and Artificial Intelligence: A Review

    All Authors Name: Anuja Bhargava et al.

    Publication year: 2021

    Objective: Review CV/AI methods for plant leaf disease detection

    Methodology: Survey of ML, DL, few-shot and molecular diagnosis from leaf images

    Key findings: Outlines strengths/weaknesses of image-based diagnosis and future directions

    Contributions: Helps position betel leaf research within broader AI disease-detection landscape

    Limitations: No direct betel experiments; limited discussion of fusion

    Relevance: Supports justification for applying advanced DL/fusion to Piper betle

    URL/DOI : 10.1109/ACCESS.2024.3373001

17. **Paper title:** PlantIF: Multimodal Semantic Interactive Fusion via Graph Learning for Plant Disease Diagnosis

    All Authors Name: Xingcai Wu; Jiawei Zhang; Ziang Zou; Chaojie Chen; Ya Yu; Peijia Yu; Yuanyuan Xiao; Qi Wang; W.M.W.W. Kandegama; Gefei Hao

    Publication year: 2025

    Objective: Design a multimodal fusion framework for plant disease diagnosis

    Methodology: Graph-learning–based multimodal semantic interactive fusion (images plus other modalities)

    Key findings: Reports improved diagnostic performance over unimodal baselines via semantic multimodal fusion

    Contribution: Introduces a modern multimodal graph-fusion architecture for plant disease tasks

    Limitations: No Piper betle data; modalities described at high level in abstract

    Relevance: Strong conceptual template for betel leaf multimodal (image + text/sensor) systems

    URL/DOI : https://doi.org/10.1016/j.plaphe.2025.100132

18. **Paper title:** Study of digital image processing techniques for leaf disease detection and classification

    All Authors Name : Gittaly Dhingra; Vinay Kumar; H. Joshi

    Publication year: 2018

    Objective: Review image-processing approaches (1997–2016) for leaf diseases

    Methodology: Classical image processing pipeline review (segmentation, features, classifiers)

Key findings: Summarizes early image-processing techniques and challenges

Contributions: Historical background for modern DL/fusion methods

Limitations: Pre-DL era; no betel; no multimodal fusion

Relevance: Informs preprocessing/feature-engineering baselines for betel datasets

URL/DOI : https://link.springer.com/article/10.1007/s11042-017-5445-8

19. **Paper title:** Enhanced Plant Disease Classification using Deep Learning and Digital Image Processing Techniques

All Authors Name: Koteswararao Yenni; K. V

Publication year: 28-05-2025

Objective: Present an improved CNN-based plant disease classifier with robust preprocessing

Methodology: CNN with denoising, contrast enhancement and augmentation, trained on five diseases (tomato leaf mold, potato blight, corn rust, apple scab, grape black rot)

Key findings: Reached 89–94% accuracy, outperforming traditional methods; supports real-time predictions

Contribution: Shows value of combined preprocessing and DL in achieving reliable, real-time disease diagnosis

Limitations: Focused on few crops; monomodal RGB; no fusion with environmental or spectral data

Relevance: Technique can be ported to betel leaf images for more robust field deployment

URL/DOI : https://doi.org/10.1109/ICCRTEE64519.2025.11053028

20. **Paper title:** Analysis of Automatic Plant Disease Classification Using Image Processing Techniques

All Authors Name: Punam; Rakesh Goyal

Publication year: 28-04-2022

Objective: Analyze automatic plant disease classification approaches using image Processing

Methodology : Review of image capture, processing and modeling strategies, including remote sensing and ML

Key findings : Summarizes benefits and drawbacks of various techniques across crops and imaging setups

Contributions: Helps identify suitable image processing pipelines and data needs for automated systems

Limitations: High-level; limited discussion of deep multimodal fusion; no betel-specific data

Relevance: Conceptual basis for designing betel leaf image pipelines

URL/DOI : https://doi.org/10.1109/ICACITE53722.2022.9823862

21. **Paper title:** Early betel leaf disease detection using vision transformer and deep learning algorithms

All Authors Name: S. Kusuma, K. Jothi

Publication year: 2023

Objective: Compare CNNs and a Vision Transformer for early betel-leaf disease Detection

Methodology: Supervised image classification using VGG19, DenseNet201,

ResNet152V2 and ViT on betel-leaf images; accuracy used as main metric

Key findings: DenseNet201 achieved 98.77% test accuracy; ResNet152V2 98.42%; ViT 97.83%; VGG19 91%

Contributions: First direct CNN vs ViT comparison on betel leaf; shows high viability of deep models for this crop

Limitations: Generalization to new fields/seasons and deployment constraints not evaluated (no field trials mentioned)

Relevance: Core benchmark for deep models (CNNs, ViT) on betel-leaf disease Classification.

URL/DOI : https://link.springer.com/article/10.1007/s41870-023-01647-3

22. **Paper title:** Image Processing Based Leaf Rot Disease, Detection of Betel Vine (Piper Betel L.)

All authors name: A. K. Dey, Manisha Sharma, M. Meshram

Publication year: 2016

Objective: Detect and measure leaf rot severity in betel vine using low-cost image processing

Methodology: Color-feature analysis, segmentation of rotten regions, area computation of rotted portion from digital images

Key findings: Vision-based system could automatically segment and quantify rot area with “promising performance”

Contributions: Early, inexpensive, farmer-oriented rot-severity measurement workflow for betel vine

Limitations: Focuses only on leaf rot; performance numbers and robustness across conditions not fully detailed

Relevance: Baseline for classical (pre-deep-learning) betel-leaf disease

Quantification

URL/DOI : https://doi.org/10.1016/j.procs.2016.05.262

23. **Paper title:** Agronomy with IoT Devices: The Smart Solution for Detection of Diseases of Betel Leaves

All Authors Name: Indranil Sarkar et al

Publication year: 2023

Objective: Integrate IoT-based smart agronomy with automated betel-leaf disease

detection using image analysis and SVM

Methodology: Betel-leaf image collection; preprocessing, ROI segmentation, feature extraction (color/texture); SVM classifier implemented within an IoT-based decision system

Key findings: Proposed system can automatically distinguish normal vs abnormal leaves; detailed quantitative metrics not fully highlighted in abstract

Contributions: Connects betel-leaf disease classification with IoT smart-farming framework

Limitations: Limited performance reporting; scalability and security challenges of IoT only discussed conceptually

Relevance: Useful for work at the intersection of IoT and betel-leaf disease monitoring.

URL/DOI: 10.13189/ujar.2023.110109

24. **Paper title:** A Leaf Disease Classification Model in Betel Vine Using Machine Learning Techniques

All Authors Name: N. Zeba, M. A. Malek, Sanjida Sultana Reya

Publication year: 2021

Objective: Classify betel-vine diseases (Bacterial Leaf Spot, Stem Leaf) using classical ML

Methodology : 1275 images captured by phone; Gaussian mixture model for segmentation; SVM classifier

Key findings: Achieved 83.69% classification accuracy, competitive with contemporary ML approaches

Contributions: Benchmarks non-deep-learning pipeline for two specific betel-vine diseases

Limitations: Small dataset; limited to two diseases; lower accuracy than modern deep models

Relevance: Comparative baseline against deep-learning approaches on same diseases.

URL/DOI : https://doi.org/10.1109/ICREST51555.2021.9331142

25. **Paper title:** An Approach to Identify Diseases in Betel Leaf Using Deep Learning Techniques

    All Authors name: Md Abdul Malek, Sarnali Basak, Sanjida Sultana Reya

    Publication year: 2022

    Objective: Compare multiple transfer-learning CNNs and traditional ML for classifying two betel-leaf diseases

    Methodology: 5800 images (stem leaf, Bacterial Leaf Spot); preprocessing with CLAHE and GMM; models: VGG16/19, ResNet50, AlexNet, InceptionV3, SVM, Logistic Regression

    Key findings: InceptionV3 achieved best test accuracy of 94.83% on 658 unseen images

    Contributions: Comprehensive model comparison for betel-leaf disease; establishes InceptionV3 as strong baseline

    Limitations: Only two diseases considered; no edge-deployment or latency analysis

    Relevance: Useful benchmark study for choosing deep-learning backbone for betel-leaf tasks.

URL/DOI: https://doi.org/10.1109/STI56238.2022.10103348

26. **Paper title:** Application Development for a Project using Flutter (betel leaf disease detection with YOLOv3)

    All Authors name: K. Nagaraj, B. Prabakaran, M. Ramkumar;

    Publication year: 2022

    Objective: Implement YOLOv3-based betel-leaf disease detector with transfer learning to support farmers via an app-like system.

    Methodology: Transfer learning on YOLOv3 for betel-leaf images; parameter tuning for speed/accuracy balance.

Key findings: Achieved ~95.6% precision in detecting betel-leaf infections .

Contributions: Moves from pure model research to near-deployment scenario (Flutter-based application concept) .

Limitations: Details on dataset size, disease types, and on-device performance are limited in abstract .

Relevance: Important for mobile/real-time betel-leaf disease detection and app development.

URL/DOI: https://doi.org/10.1109/ICOSEC54921.2022.9951938